



Fiona Stanley Fremantle Hospitals Group

Policy and Procedure

Recognising and Responding to Acute Deterioration

Reference #: FSFHG-HW-POL-0033

Scope

Site	Service/Department/Unit	Disciplines
Fiona Stanley Hospital Fremantle Hospital	Hospital Wide	All Staff

1. Introduction

Serious adverse events such as unexpected death and cardiac arrest are often preceded by observable physiological and clinical abnormalities. Early recognition of acute deterioration, followed by prompt and effective action, may minimise the occurrence of adverse events, such as cardiac arrest, and may mean a lower level of intervention is required to stabilise a patient. The Fiona Stanley Fremantle Hospital Group (FSFHG) has systems in place to support and promote recognition of and response to patients whose condition deteriorates.

The purpose of this policy is to:

- Ensure that safe, early and appropriate management of a medical emergency, including cardiopulmonary resuscitation occurs at FSFHG
- Detail the duties and training requirements for all staff within FSFHG relating to the management of a medical emergency including cardiopulmonary resuscitation
- Standardise the management of medical emergencies and cardiac arrests within FSFHG services in accordance with the current Australian Resuscitation Council guidelines.

This document is to be read in conjunction with the “FSH - Code Blue”, the “Fremantle Hospital Code Blue - Medical Emergency” or “FSH - Cockburn Health Code Blue – Medical Emergency” procedures from the applicable site Emergency Procedure Manuals.

The Code Blue Medical Emergency chapter of the Emergency Procedure Manuals contains Action Cards that list the specific roles and responsibilities of Medical

Emergency Team (MET) members and local clinicians responding to a Code Blue medical emergency.

NOTE: There are significant variations in Medical Emergency Team membership, processes, equipment and resources at the Cockburn Health site, please see [Appendix 10](#) for site specific MET related information.

2. Terminology

MET	<u>FSH</u> Medical Emergency Team(s): The FSH response to medical emergencies occurring within the MET boundary, comprises of the following teams: • Adult: Responding to medical emergencies concerning adult patients. • Neonate: < 28 days • Paediatric: From 28 days to 16 years of age • Obstetric: Responding to obstetric emergencies Note: MET activation can occur for visitors/staff and outpatients attending FSFHG. FSH DOES NOT provide a MET response for visitors/staff and outpatients located external to Medihotel ward M6 Medihotel ward M7 or the L2 Medihotel Link Bridge. <u>FH</u> Medical Emergency Team(s): The FH response to medical emergencies occurring within the MET boundary, comprises of the following teams: • Adult: Responding to medical emergencies concerning adult patients and adolescents >14 to 16 years old. • Paediatric: Responding to unplanned public Paediatric presentations: From 0 days to 14 years of age Note: MET activation can occur for visitors/staff and outpatients attending FSFHG <u>FSH -Cockburn Health</u> Medical Emergency Team: Cockburn Health responds to
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	medical emergencies occurring within the MET boundary
ART	Advanced Response Team: The FSH response to medical emergencies occurring outside of the MET boundary.
SIMS	The Safety and Incident Management Service at FSH
Code Blue	A medical emergency.
RTLS	Real Time Location System – Duress device specifically for staff at FSH-CH.

3. Responsibilities

3.1. Recognition and response to Acute Deterioration (RRAD) governance committee

- The RRAD Governance Committee facilitates the governance and monitoring of the acute deterioration, recognition and response systems and processes across FSFHG in accordance with the requirements of the National Safety and Quality Health Service (NSQHS) Standards.

3.2. Assessment Liaison Escalation Response Team (ALERT)

- The ALERT Service is a multidisciplinary team intended to provide care across Fiona Stanley and Fremantle hospitals, 24/7. While it does work across both sites, the underlying ethos and structure remain the same and can be broadly divided into a proactive consult service and reactive emergency response service.
- The proactive consult service involves actively rounding on wards searching for acutely unwell or deteriorating patients as identified by ward nursing or medical staff. ALERT also provide direct assistance to home team who may be struggling to manage deteriorating patients, particularly those procedural specialities where team Registrars may not be immediately available to assist. The reactive component consists of MET involvement and the completion of usual after-hours roles and ward cover duties by residents and/or registrars.
- At FSH the ALERT team is divided into the general rounding team and the inbound admissions unit (IAU) team. The IAU registrar at FSH covers areas of the hospital where patients are likely to arrive from the Emergency Department or from other hospitals via transfer and includes the Satellite Acute Medical Unit (SAMU), Transit Lounge 1 and 2 (T1 and T2), the Acute Medical Unit (AMU) and Acute Surgical Unit (ASU).
- The Rounding Registrar covers most other wards of the hospital excluding

paediatrics, obstetrics, and birthing suites. Of note, the Medihotel, State Rehabilitation Service (SRS), Mental Health Unit (MHU) and Coronary Care Unit (CCU) should form part of this round.

- RMO 'Medihotel' covers Medihotel wards M6 and M7 as well as DSU / SSU
- At FH, The ALERT Registrar will round with the ALERT CNS. Throughout these shifts the ALERT Reg/CNS will Attend any concurrent MET as the Team Leader and Primary MET activations as a consultation service and proactively round all clinical areas (including W block) throughout their shift to provide support, advice, and clinical expertise to ward staff caring for 'at risk' and deteriorating patients.

3.3. Medical Emergency Team (MET) coordinator

- The MET Coordinator is responsible for the coordination and facilitation of the Medical Emergency Team (MET) service for FSFHG ensuring the provision of policy, procedures, documentation and evaluation associated with recognising and responding to acute deterioration and resuscitation.
- The role also leads quality improvement and research to support the development of best practice initiatives and training for RRAD.
- The MET Coordinator manages the MET SRN rotating roster and responds to MET activations (FSH and FH ONLY).
- The MET Coordinator reviews all METs that occur from 2300 - 0800Hrs Mon – Fri to ensure that all interventions are completed and that an appropriate post MET management plan is in place for the patient (FSH and FH ONLY).

3.4. Medical Emergency Team (MET)

- The MET is responsible for responding within agreed timelines to calls to assess and stabilise the acutely deteriorating patient by providing the required therapeutic intervention. The MET makes decisions regarding the need to escalate care and oversees safe intra-hospital transfer of the patient where required. The MET is also responsible for informing the patient's treating team of interventions undertaken.

3.5. Nurse/midwifery unit managers and department managers

- Unit and Department Managers are responsible for ensuring that staff in their area complete the necessary training as outlined in this policy and that they inform staff of any changes in practice. They are responsible for ensuring that their resuscitation trolley is checked weekly and after use with the required equipment available and that they escalate any areas of concern regarding resuscitation to the MET Coordinator.

3.6. All clinical staff

- All clinical staff are responsible for complying with FSFHG policy and

procedure regarding management of clinical deterioration. Clinical staff are expected to evaluate a patient's condition when taking observations, triggering a MET response for deteriorating patients when appropriate, initiating appropriate interventions prior to arrival of the MET and supporting patient's ongoing management as per the Australian Resuscitation Guidelines (see Appendices 1, 3, 4 & 5).

4. Orientation, training and education requirements

- All clinical staff should be trained and annually assessed for competency in Life Support in accordance with their role as per the [SMHS Safety Skills Training Policy](#) and the FSFHG Hospital Life Support Training Framework.
- All casual and agency staff are orientated to new clinical areas using the relevant Unit Orientation Information Sheet and Checklist.
- To support staff in management of the deteriorating patient and hospital life support, a number of resources are available including the [FSFHG Adult Hospital Life Support Resource](#), [FSFHG Paediatric Hospital Life Support Resource](#) and the [FSH Medical Staff Orientation Manual](#).

To ensure a timely well-coordinated response to medical emergencies, clinicians must familiarise themselves with the roles and responsibilities listed on MET Action Cards (AC). In addition, local processes to support MET must be identified and understood, including way finding to assist MET on arrival, resuscitation trolley layout, and the location of additional equipment of potential use (e.g. IV trolley).

5. Patient monitoring and observations

- Monitoring of patient vital signs is an essential component of caring for patients in order to assess for effectiveness of treatments, detect procedural complications and identify early signs of acute deterioration.
- The FSFHG Observation and Response Chart Policy outlines the minimum expectations regarding the monitoring of patients within FSH and FH.
- The FSFHG Observation and Response Chart Policy assists staff to recognise and respond to an early warning indicator of acute deterioration. Staff must escalate clinical deterioration as per the relevant observation and response chart.

5.1. Assessing and managing the deteriorating patient

- All clinical staff must be able to systematically assess a conscious, clinically deteriorating patient who may require activation of the MET and initiate appropriate interventions according to their professional practice and

training using the Airway, Breathing, Circulation, Disability, Exposure (ABCDE) approach.

- On finding an unconscious person, staff are to follow the Basic Life Support (BLS) Algorithm/Flow Chart which supports the preservation or restoration of life by the establishment of and/or the maintenance of airway, breathing, circulation and related emergency care. This includes the use of Automated External Defibrillators (AED) ([see Appendix 1](#)).
- Refer to the “ALERT – Assessment, Liaison Escalation Response Team: Departmental orientation guide for medical staff” for specific management approaches for mental health and surgical patients.
- When evidence of acute deterioration occurs or staff are concerned about a patient’s condition, they should escalate care as outlined in the relevant observation and response chart and the FSFHG “Escalation Process for Nursing and Midwifery Staff” policy. When escalating care or activating the MET, nursing staff should handover the patient using the iSoBAR format as outlined in the FSFHG “Clinical Handover” policy.

6. MET and Care call essentials

- To activate a code blue MET phone “55”, state “Code Blue Medical Emergency”, and provide the following information:
 - the type of medical emergency (E.g. Adult, Neonate, Paediatric, Obstetric)
 - the exact location and
 - the room number.
- FH staff: see [Appendix 2](#) for FH Adult MET Activation Flowchart.
- FSH - Cockburn Health staff: see [Appendix 10](#) for FSH - Cockburn Health MET Activation process.
- The criteria for triggering an adult MET response are outlined in Table 1
- The criteria for triggering a paediatric MET response are outlined in [Table 2](#)
- See [Appendix 3](#) for the Australian Resuscitation Council’s Advanced Life Support Algorithm

Table 1. Adult MET Criteria

Airway	Threatened
Breathing	Respiratory rate ≤ 4 or > 36 breaths per minute Oxygen Saturation $\leq 84\%$

Circulation	Heart rate < 40 or > 140 beats per minute Systolic blood pressure < 90mmHg
Neurology	Sudden fall in level of consciousness (Glasgow Coma Scale, GCS) of ≥ 2 points Repeated or prolonged seizures
Other	Any person who you are seriously concerned about that does not meet the above criteria.

MET activation occurs based upon single parameter or total ADDS score meeting set criteria. The value used to calculate the Systolic blood pressure (SBP) and Heart rate (HR) should be measured in a supine patient. Physiological response to posture may be significant in some cases and the patient should be laid back down and SBP and HR repeated, within 5 minutes. A MET may then be triggered based upon the subsequent supine SBP or HR (and/or the subsequent total ADDS).

As always, if there is ongoing nursing concern without a total score or single parameter trigger then a MET should be activated

6.1. Exceptions to adult Code Blue Medical Emergency criteria:

6.1.1. Management of Cardiac and Thoracic Surgery patients on Ward 4C

- If in Atrial Fibrillation (AF) or Atrial Flutter (A. Flutter) with the following parameters, no MET needs to be activated
- HR less than 180bpm
- no other physiological parameters meeting MET criteria
- An ECG is to be performed to confirm AF or A. Flutter, ruling out all other tachycardias inclusive of A. Flutter
- The Cardiothoracic medical team are on site and able to attend immediately
- In the event that the AF is not managed/reverted in a timely fashion, or the Cardiothoracic medical team are not onsite/unavailable, a MET must be activated.
- Refer to the FSFHG Cardiac Surgery and Thoracic surgery policies for the local escalation protocol for cardiothoracic patients on Ward 4C

6.1.2. Cardiovascular and respiratory outpatient gyms

- The Adult Observation Response Chart (AORC) chart is designed to detect abnormal physiological response to illness and thus deterioration at rest; and has less utility to detect deterioration in patients undergoing exercise
- A full set of baseline observations should be recorded prior to exercise commencing.
- Patients who have observations that meet MET criteria at rest (which may be normal for them), require modifications to be recorded in the digital medical record (DMR) by their medical consultant.
- During exercise:
 - If the patient is asymptomatic and under the supervision of a Physiotherapist (P) or Exercise Physiologist (EP) a deterioration of physiological parameters does not need to trigger a MET response.
 - If the patient becomes symptomatic (e.g., chest tightness, uncharacteristic breathlessness, pre syncope), the exercise should immediately be ceased, and observations performed at a minimum of 5 minutely until resolution of symptoms. If the symptoms resolve and the observations return to baseline, a MET response is not required.
- If the patient has ongoing symptoms despite cessation of exercise, then urgent medical review is required, by the home team. Any significant delay to this review should trigger a MET response.

- If at any time the P/EP is concerned about the patient – a MET must be activated.

Table 2 Paediatric MET Criteria

	< 3 months	3 – 12 months	1-4 years	5-11 years	>12 years
Airway	Airway threat or Apnoea	Airway threat or Apnoea	Airway threat or Apnoea	Airway threat or Apnoea	Airway threat or Apnoea
Breathing	Respiratory Rate <20 breaths/min	Respiratory Rate <15 breaths/min	Respiratory Rate <10 breaths/min	Respiratory Rate <5 breaths/min	Respiratory Rate <5 breaths/min
Circulation	Heart rate ≤ 70beats/min Systolic blood pressure ≤ 40 mmHg	Heart rate ≤ 70beats/min Systolic blood pressure ≤ 50 mmHg	Heart rate ≤ 60beats/min Systolic blood pressure ≤ 50 mmHg	Heart rate ≤ 50beats/min Systolic blood pressure ≤ 50 mmHg	Heart rate ≤ 40beats/min Systolic blood pressure ≤ 60 mmHg
Neurology	Repeated or prolonged seizures Sudden fall in level of consciousness	Repeated or prolonged seizures Sudden fall in level of consciousness	Repeated or prolonged seizures Sudden fall in level of consciousness	Repeated or prolonged seizures Sudden fall in level of consciousness	Repeated or prolonged seizures Sudden fall in level of consciousness
Other	Any patient who you are seriously concerned about that does not fit the above criteria	Any patient who you are seriously concerned about that does not fit the above criteria	Any patient who you are seriously concerned about that does not fit the above criteria	Any patient who you are seriously concerned about that does not fit the above criteria	Any patient who you are seriously concerned about that does not fit the above criteria

6.1.3. Refer to [Appendix 4](#) for the management of Paediatric Cardiorespiratory Arrest

6.2. Recognition and response for the deteriorating neonate (FSH Only).

- The response and escalation criteria and required clinician actions to activate each level of review and escalation are outlined in the tables in [Appendix 5](#) and are to be used in conjunction with the Australian and New Zealand Committee on Resuscitation (ANZCOR) Newborn Life Support algorithm ([Appendix 6](#))

A Code Blue Neonate AND Code Blue Paediatric MUST both be activated for any Neonates/Newborns < 28 days old presenting to The Emergency Department or admitted on Ward 3A.

- Each MET member is equipped with a paging device which will alert on MET activation. At FSH, MET members must acknowledge (via the device) that a MET message has been received. MET pagers that are role dependent must be handed over to the appropriate clinician who assumes the MET role. Refer to the FSH MET Activation in a Paging Outage Standard Operating Procedure and the FSH Code Yellow Paging Disruption - Standard Operating Procedure for **procedures to ensure continuity of MET responses** in the event of paging downtime.
- To ensure appropriate situational awareness of hospital wide MET activity, additional clinicians not rostered to MET activations are also informed (via paging devices) of Code Blue calls (i.e. certain anaesthetic clinicians, Hospital-Out-of-Hours staff).
- A MET call can only be stood down by the MET leader in discussion with the MET following a review of the patient/situation, and where the patient is no longer in MET criteria. The MET Lead must formulate a plan for the patient upon stand-down of the MET; communicate this to the ward team and document this in the patient's integrated progress notes as well as the Medical Emergency Record.
- Where appropriate, modifications to the physiological parameters for MET activation are to be documented on the relevant observation and response chart and the medical record in accordance with the requirements of the FSFHG Observation and Response Chart.
- Prior to stand down of MET, the patient's relevant physiological parameters and escalation score need to be taken into consideration by the team, in the context of patient safety and disposition. Consideration to be given to ability of ward to provide increased physiological monitoring and care.
- At time of stand down of MET call, a set of observations must be transcribed onto the relevant observation and response chart. Observations must then be performed 15 minutely for 30 minutes, then 30 minutely for one hour unless otherwise indicated on the relevant observation and response chart.

6.3. Aishwarya's CARE Call – Call and Respond Early (CARE) patient and carer activated escalation

- Aishwarya's CARE escalation process allows the patient, relative or carer to raise their concerns when the patient's clinical condition is deteriorating, and they feel that an appropriate intervention has not occurred.
- The CARE call process enhances communication between the patient/relative/carers and the health care team to optimise health outcomes and embed best practice.
- Patients, relatives and carers should be informed of the process on

admission as per the FSFHG Aishwarya's CARE Call – (Call and Respond Early): Patient and Carer Activated Escalation policy

6.4. Public presentations at Fremantle Hospital

- At FH, members of the public may present to the hospital requiring access to an Emergency Department or medical assistance.
- If members of the public present to FH with a life-threatening situation that falls within the MET criteria, a MET call is placed and actions taken as outlined in the FSFHG Fremantle Hospital - Requests from the Public for Medical Attention guideline.
- For unplanned public paediatric presentations ALERT CNS must be contacted, and paediatric grab bag is available in the B3 ALERT office.

6.5. Medical Emergency Team (FSH and FH)

- The MET is comprised of the following members of staff:
 - MET Doctor 1 ICU Registrar (FSH)/Medical Registrar (FH) AC 1
 - MET Doctor 2 Airway (FSH)/Perioperative Care Unit RMO (FH) AC 2
 - MET Doctor 3 Medical Liaison (FSH)/Anaesthetic Registrar (FH) AC 3
 - MET Doctor 4 RMO/Intern (FSH)/Medical RMO (FH) AC 4
 - MET Senior Nurse (FSH)/Senior Registered Nurse (FH) AC 5
 - MET Porter (FSH)/Patient Service Assistant (FH) AC 6
- The MET Team Lead for the day must be allocated at the beginning of the morning MET huddle. The huddle occurs at FSH in the ALERT Office and at FH in the Medical Handover Room.
- All resuscitation attempts should have a team leader. The team leader should identify themselves immediately when attending the MET call and confirm their role as team leader.
- The ICU Registrar (FSH) will carry a MET bag to all MET activations. For bag contents list and re-stocking procedure see Appendices [Appendix 7](#) and [Appendix 9](#). This bag is checked by the rostered ICU MET Registrar daily.
- The Primary MET SRN across FSFHG will carry a GRAB bag to all MET activations, it is the responsibility of the rostered MET SRN to check this bag daily. For bag contents list see [Appendix 8](#)
 - For FSH bag re-stocking procedure see [Appendix 9](#)
 - At FH the drug contents can be obtained from the 8N/PCU medication imprest
- For specific roles and responsibilities of the FSH MET members see [FSH Emergency Procedures - Code Blue Medical Emergency](#). For FH see [FH Emergency Procedures - Code Blue Medical Emergency](#).

- Note: At FH, the airway specialist overnight and on weekends is the Consultant Anaesthetist (On-Call)
- See [Appendix 10](#) for FSH - Cockburn Health MET membership

6.6. Concurrent Code Blue MET activations

- At FSH, a secondary MET team is rostered for activation in the event of concurrent MET calls. The process for activating the Adult Secondary MET is as follows:
 - Only the Primary MET SRN can activate the secondary Adult MET team
 - The decision to activate the secondary Adult MET team is made by the MET SRN and MET Team leader.
 - The team leader will dial '55' or direct the MET senior nurse to dial '55' and state 'Activate the Secondary Adult Code Blue Medical Emergency Team', providing the exact location and room number.
 - On receiving this call, Helpdesk will activate the secondary Adult MET team stating in the message Code Blue Medical Emergency and providing the location and room number.
 - The primary MET team are to attend and relieve the secondary Adult MET team as soon as practicable.

6.7. Escalation process when ICU unable to attend MET calls (FSH)

- ICU Duty Consultant (DC) to notify the Director of Clinical Services (DCS) in hours or On-Call Medical Director (After Hours)
- ICU DC to state reason for withdrawal of ICU MET member, for example: ICU medical staffing profile or ICU patient acuity etc.
- ICU DC to state the duration of projected withdrawal period, for example: number of hours or whole shift.
- An ICU Registrar must attend any MET when patient management/acuity requires it as requested by any member of the MET team.
- An ICU Registrar must remain contactable on Ext: 28801 for the duration of the ICU withdrawal period.
- In hours, the ICU DC must notify the ALERT CNS on Ext: 27654 and MET Medical Registrar on Ext: 28889
- After hours, the Medical Director On-Call must notify the Hospital Coordinator on Ext: 28168 who will notify the Hospital-Out-Of-Hours Team (HOOT) CNM's and ALERT CNS.
- ALERT and HOOT SRN to ensure that all delegated MET members for duration of ICU withdrawal period are aware of change in process and alternative escalation to ICU process.

- In the event of an ICU Registrar not participating in the MET the General Medicine Medical Registrar or Anaesthetist may assume the role of MET Doctor 1/Team Leader and carry the MET bag as delegated.

6.8. Escalation process when Anaesthetics unable to attend MET calls (FSH)

- Duty anaesthetist (DA) to notify the Director of Clinical Services (DCS) in hours or On-Call Medical Director (After Hours)
- DA to state reason for withdrawal of anaesthetic Registrar as MET member, for example: staffing profile etc.
- DA to state the duration of projected withdrawal period, for example: number of hours or whole shift.
- An Anaesthetic Registrar must attend any MET when patient management/acuity requires it as requested by any member of the MET team.
- An Anaesthetic Registrar must remain contactable on Ext: 28802 for the duration of the withdrawal period.
- After hours, the Medical Director On-Call must notify the Hospital Coordinator on Ext: 28168 who will notify the Hospital-Out-Of-Hours Team (HOOT) CNM's and ALERT CNS, to ensure that all MET members are aware of the temporary change in process.
- ALERT CNS and HOOT SRN are to ensure that all delegated MET members for duration of the Anaesthetics withdrawal period are aware of change in process and alternative escalation to Anaesthetics process.

6.9. Escalation process when Acute Medical Unit (AMU) Registrar unable to attend secondary MET calls (FSH)

- AMU Consultant to notify the Director of Clinical Services (DCS) in hours or On-Call Medical Director (After Hours)
- AMU Consultant to state reason for withdrawal of AMU Registrar as MET member, for example: staffing profile etc.
- AMU Consultant to state the duration of projected withdrawal period, for example: number of hours or whole shift.
- AMU Consultant to liaise with ALERT Registrar (Ext: 27780) to determine appropriate replacement Registrar for duration of withdrawal period.
- After hours, the Medical Director On-Call must notify the Hospital Coordinator on Ext: 28168 who will notify the Hospital-Out-Of-Hours Team (HOOT) CNM's and ALERT CNS, to ensure that all MET members are aware of the temporary change in process.
- ALERT and HOOT SRN are to ensure that all delegated MET members for duration of the Anaesthetics withdrawal period are aware of change in

process and alternative escalation to Anaesthetics process.

- If there is no ALERT Inbound Admissions Unit (IAU) Registrar to carry the Secondary MET pager (In-Hours), this is to be carried by the dedicated AMU Registrar.

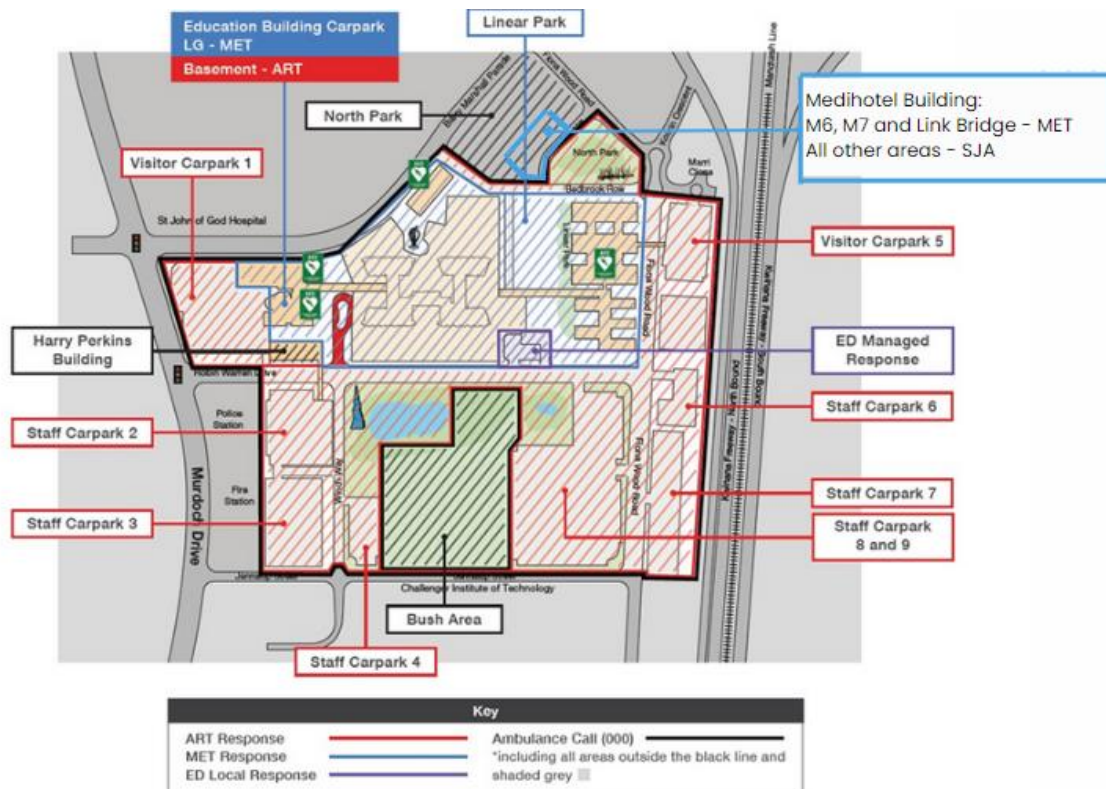
6.10. MET and Advanced Response Team (ART) boundaries

- Advanced Response Team (ART) processes have been established to effectively manage code blue medical emergencies that occur on the grounds of the hospital. To activate an ART response, calls are made as usual and when the location is given switchboard/helpdesk activate the ART response.

6.10.1. At FSH

- MET boundaries include the buildings and open space at the western entrance and the LG level car park under the education building, which is also the oncology drop off area Including the Medihotel building (Wards M6 and M7, inclusive of the link bridge extending from the theatre corridor to the Medihotel service lifts).
- A full resuscitation trolley is located in the PathWest specimen collection area and in the Allied Health 2 on the ground floor, which is accessible by a porter, to enable support to non-clinical areas.
- St John's Ambulance (SJA) and Safety and Incident Management Service (SIMS) will be the initial ART outside the MET boundary. The additional Emergency Department medical ART response is available on request by SIMS, SJA or any hospital staff member.
- The ART boundaries which SIMS and ED attend (if requested) include the on-site carparks, ambulance area immediately adjacent to ED and the basement car park one floor down under the education building.
- The Harry Perkins building is not part of FSH, therefore no hospital response is sent, ambulance should be called on '000' (see [Figure 1](#)).
- The Medihotel Piazza, Ground level, Level 1, Level 2, Level 3 and Level 4 are not part of FSH, therefore no hospital response is sent, ambulance should be called on '000' (see [Figure 1](#)).

Figure 1: FSH MET and ART Boundaries



6.10.2. At FH

- The MET will attend all Fremantle Hospital buildings and their entrances, adjoining pathways and courtyards. This includes Blocks A, B, D, E, F, L, M, N, O, P, T, V and W, all on-site car parks and adjacent footpaths directly bordering the hospital.
- The FH boundary is defined by South Terrace, Alma Street, Hampton Road, Fothergill Street and the Fremantle Oval.
- All other areas external to the FH boundary are unable to receive a MET response. First responders should provide assistance where able, or call an ambulance if necessary, on '000' (See [Figure 2](#)).

Figure 2: FH MET Boundaries



7. Resuscitation equipment

- Resuscitation equipment requirements are outlined in the FSFHG “Resuscitation and Emergency Equipment Content Checking and Layout (Adult)” policy, the “Paediatric and Neonatal resuscitation and emergency equipment layout and contents checking” policy.

7.1. Automated External Defibrillators (AEDs)

- There are several Automated External Defibrillators (AED) located in non-clinical areas at Fiona Stanley Hospital and Fremantle Hospital for use in emergency situations, locations are in Table 3 and [Table 4](#).

Table 3 Fiona Stanley Hospital AED Locations

Building	Location
Administration	Ground floor behind the reception desk in the main foyer.
Education	Ground floor behind the main reception desk.
Pathology	Second floor at the east end, behind the PathWest management reception (room no AL20135).
State Rehabilitation Service	Lower ground floor in the hydrotherapy pool first aid room.

Table 4: Fremantle Hospital AED Locations

Building	Location
B Block, level 3	Next to the reception desk.
M Block, level 6	Opposite the security office.

7.2. Patients with an advanced health directive or not for CPR

- A Not for Cardiopulmonary Resuscitation (NFCPR) decision does not imply withdrawal of all treatment. In these situations, all patients must be monitored using an Adult Observation and Response Chart and the MET must be activated for criteria other than the cardiopulmonary arrest.
- See the FSFHG Advanced Health Directives and Enduring Power of Guardianship Policy and the FSFHG Goals of Patient Care Policy

7.3. Emergency call bell testing

- Red 'emergency' call bells are a way for nursing staff to call for urgent assistance when a sudden deterioration in the patient's condition occurs.
- Bedside and fixed emergency call bells in all clinical areas are to be formally tested annually and the results submitted to the RRAD Committee for review.
- Any faults discovered during emergency call bell testing are to be escalated to the Estates team by raising a work order.

8. Governance of MET processes

- The FSFHG RRAD Committee provides overarching governance of the recognition and response to clinical deterioration systems across the group.
- On the completion of a Code Blue medical emergency, the senior nurse on the MET is responsible for ensuring the accurate completion of the MET Record Form.
 - A copy of the form is scanned and sent to the FSH Safety Quality and Risk Unit (via the fshsqr.MET@health.wa.gov.au email address) and to the NUM of the area.
 - The original form is retained in the patient's medical record.
- MET call documentation is reviewed by the MET Coordinator to identify any failures in the recognition and response system as outlined below.

- A Quarterly report is pulled from SHIMs which identifies all MET calls resulting in an escalation of patient care. This includes:
 - Transfer to higher acuity ward
 - Transfer to another hospital
 - Death
- The MET Coordinator reviews all the cases resulting in an escalation of patient care in order to identify any issues such as failure to escalate, insufficient monitoring, missed opportunity for end of life care.
- Any cases that are identified as demonstrating issues are reported via a retrospective Datix Clinical Incident Management System (CIMS) submission. Cases highlighting positive escalation and excellent care are reported to the NUMs for dissemination to ward staff.
- The MET Coordinator meets with areas recognised as having a trend in identified issues over a three-month period. These areas are requested to submit an individual action plan to the RRAD committee to address and correct the identified trend of issues and prevent them from re-occurring.
- MET Escalation Reports are tabled at the RRAD Committee meeting on a quarterly basis for discussion and actioning.

9. Compliance/performance monitoring

- The FSFHG RRAD Committee will complete regular review of clinical incident trended data via the Datix Clinical Incident Management System will support the monitoring and effectiveness of the medical emergency response system and associated management plans.
- The FSFHG RRAD Committee will undertake regular audits and quality improvement activities as outlined on the Standard Audit Planner, including gathering feedback from staff and patients/carers on the RRAD/CARE Call systems and feeding information back to staff regarding performance of the RRAD systems.
- The FSFHG RRAD Committee is responsible for reporting and monitoring compliance with this policy to the Clinical Governance Committee

10. Related policy documents

- WA Health:
 - [WA Health Recognising and Responding to Acute Deterioration \(MP0086/18\)](#)

- SMHS:
 - SMHS Recognition and Response to Acute Deterioration (RRAD) Policy (SMHS: 9)
 - [SMHS Safety Skills Training Policy](#)
- FSFHG:
 - Advance Health Directive and Enduring Power of Guardianship Policy and Procedure (FSFH-HW-POL-0031).
 - Goals of patient care (FSF-HW-POL-0053)
 - Fremantle Hospital Code Blue - Medical Emergency
 - Fremantle Hospital - requests from the public for medical attention (FSFH-HW-GUI-0015).
 - FSH - Code Blue (FSH-SIM-PRO-0033).
 - Call and Respond Early (CARE) Patient and Carer Activated Escalation (FSFH-HW-POL-0002).
 - Observation and Response Chart (FSFH-HW-POL-0032).
 - Multiple Medical Emergency Team calls at FSH (FSFH-HW-PRO-0011).
 - Escalation Process for Nursing Staff Policy (FSFH-HW-POL-0014).
 - Clinical Handover Policy (FSFH-HW-POL-0035).
 - Resuscitation and Emergency Equipment content Checking and Layout (Adult) (FSFH-HW-GUI-0010).
 - Cardiac Surgery (FSFH-CAR-POL-0005)
 - Thoracic Surgery
- Other organisations
 - ANZCOR [Guideline 13.1 - Introduction to Resuscitation of the Newborn](#)

11. Other related documents

- ALERT – Assessment, Liaison Escalation Response Team: Departmental orientation guide for medical staff

12. Related standards

- NSQHS Standards:
 - Clinical Governance
 - Comprehensive Care

- Communicating for Safety
- Recognising and Responding to Acute Deterioration.

13. Bibliography

- [Australian Resuscitation Council Guidelines](#), January 2016
- King Edward Memorial Hospital Clinical Guidelines

14. Authorisation

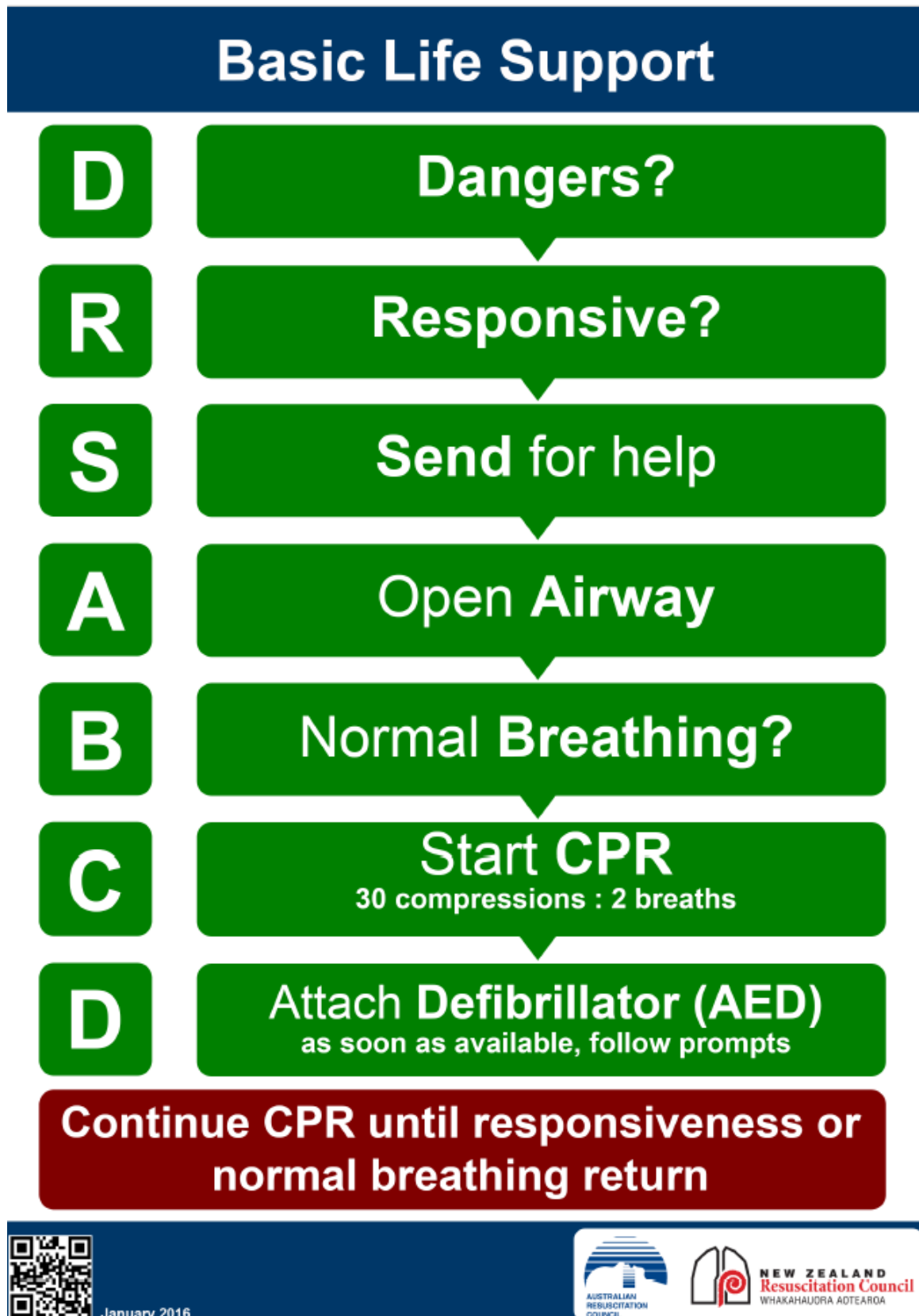
EXECUTIVE SPONSOR: Medical Director, Service 4					
Version	Date Issued	Compiled/Revised By	Committee/Consumer Group Consulted	Endorsed By	Revision due
1	09/2014	RRCD Working Group	RRCD	-	09/2016
2	05/2016	RRCD Working Group	RRCD	-	07/2017
3	10/2018	RRCD Working Group	RRCD & NMPGC	FSFHG Policy Committee	10/2019
3.1	11/2018	Nurse Director CAS	RRCD & NMPGC	FSFHG Policy Committee	11/2019
3.2	11/2018	Nurse Director CAS	RRCD & NMPGC	FSFHG Policy Committee	11/2019
4	04/2020	RRAD Committee	RRAD Committee	FSFHG Policy Committee	12/2020
5	05/2020	Nurse Director CAS	RRAD Committee	FSFHG Policy Committee	11/2020
6	07/2021	Nurse Director-Patient Flow	RRAD Committee	FSFHG Policy Committee	07/2025
6.2	02/2022	Nurse Director-Patient Flow	RRAD Committee	FSFHG Policy Committee	02/2024
6.3	03/2022	CNS MET Team	RRAD Committee NMPGC	FSFHG Policy Committee	02/2024
7	11/2023	CNS MET Team	RRAD Committee NMPGC	FSFHG Policy Committee	10/2025
7.1	11/2023; 12.2023	CNS MET Team	RRAD Committee NMPGC	FSFHG Policy Committee	10/2025

Recognising and Responding to Acute Deterioration

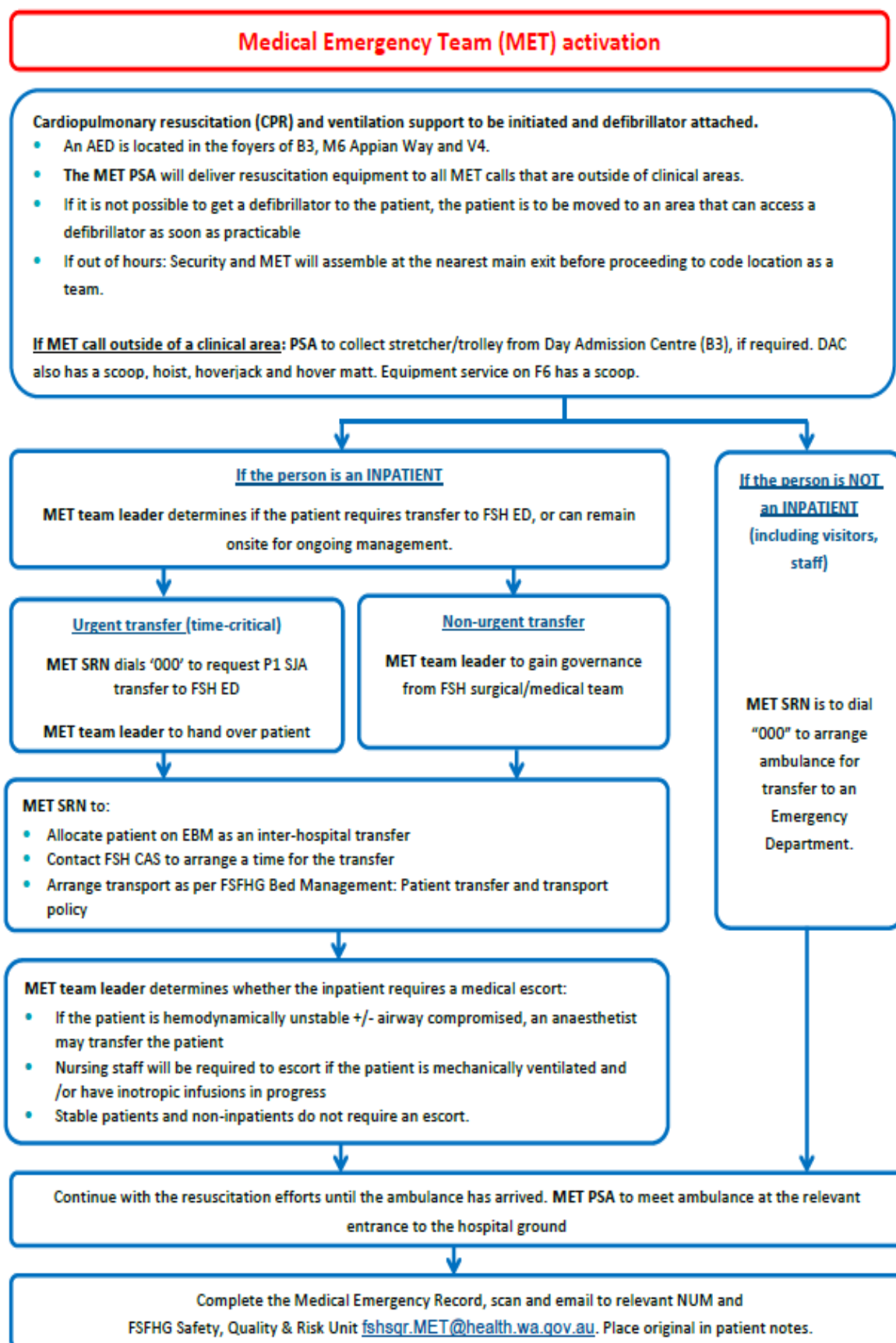
8	09/2024	MET Coordinator	RRAD Committee	FSFHG Policy committee	10/2025
8.3	10/2024	NUM – CTS AND MET Coordinator	Minor update, No Change to Review Date		10/2025
9	01/2025	NUM – CTS AND MET Coordinator	Minor update, No Change to Review Date	FSFHG Policy committee	10/2025

15. Appendices

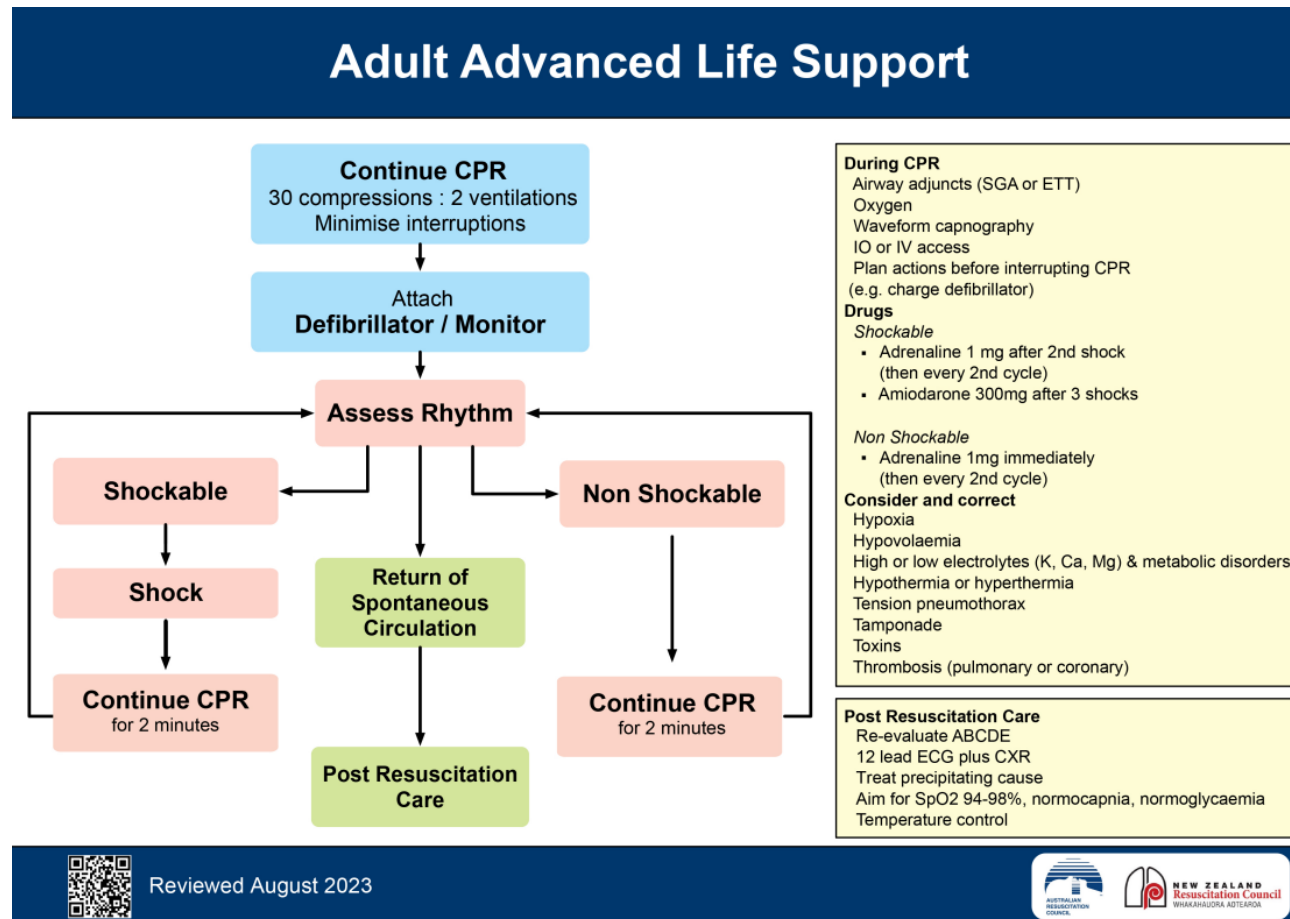
15.1. Appendix 1 ANZCOR basic life support flowchart



15.2. Appendix 2: FH MET activation flowchart



15.3. Appendix 3: Australian Resuscitation Council advanced life support algorithm: Adult cardiorespiratory arrest flowchart

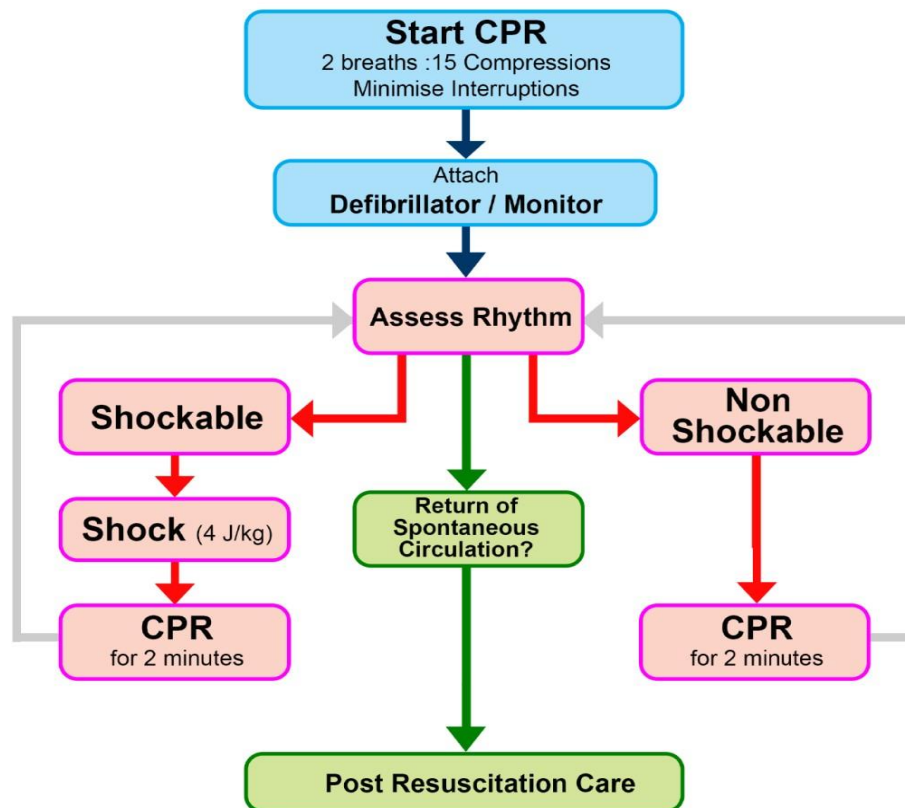


In Pregnancy:

- Manually displace the uterus to the left to remove Aortocaval compression. Add left lateral tilt, if possible, between 15 and 30 degrees.
- Consider preparation for emergency Caesarean section, as the foetus will need to be delivered if initial resuscitation efforts fail.

15.4. Appendix 4 Paediatric cardiorespiratory arrest flowchart

Advanced Life Support for Infants and Children



During CPR

Airway adjuncts (LMA / ETT)
Oxygen
Waveform capnography
IV / IO access
Plan actions before interrupting compressions
(e.g. charge manual defibrillator to 4 J/kg)

Drugs

Shockable

- * Adrenaline 10 mcg/kg after 2nd shock (then every 2nd loop)
- * Amiodarone 5mg/kg after 3 shocks

Non Shockable

- * Adrenaline 10 mcg/kg immediately (then every 2nd loop)

Consider and Correct

Hypoxia
Hypovolaemia
Hyper / hypokalaemia / metabolic disorders
Hypothermia / hyperthermia
Tension pneumothorax
Tamponade
Toxins
Thrombosis (pulmonary / coronary)

Post Resuscitation Care

Re-evaluate ABCDE
12 lead ECG
Treat precipitating causes
Re-evaluate oxygenation and ventilation
Targeted Temperature Management



January 2016



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Resuscitation Council
WHAKAHAUORA AOTEAROA

15.5. Appendix 5: Recognition and response guide for the deteriorating neonate

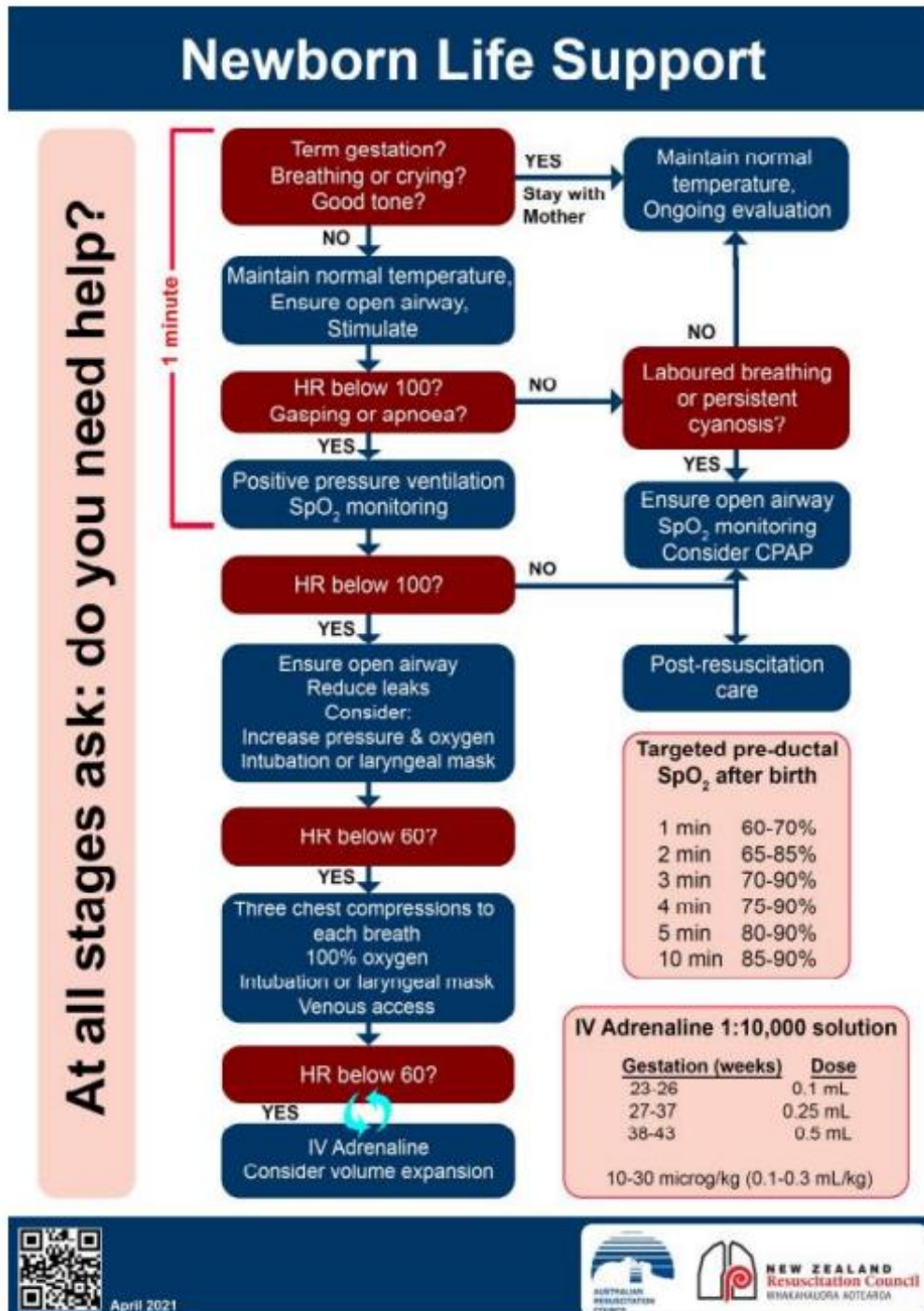
Use the tables below to determine the actions required to initiate the appropriate level of clinical review for the observed patient deterioration response

Increased Surveillance	
Deterioration Response Criteria - Changing observations - You (or a family member/carer) are worried about the infant	Actions Required - Inform Shift Coordinator (29458) - Carry out appropriate interventions as prescribed - Record observations at least every 2 hours - Monitor oxygen requirement - Manage fever, pain, fluids, distress - Consider blood gas
Shift Coordinator Review	
Deterioration Response Criteria - Instability characterised by rising FiO₂, more significant apnoea/bradycardia/desaturations episodes, rising or falling blood pressure, temperature instability, feed intolerance, lethargy or irritability, or blood results which are significantly changed or unexpected - You (or a family member/carer) are worried about the infant but they do not meet any of the above the above criteria	Actions Required - Shift Coordinator (phone 29458) must review patient - Record observations at least every hour - Monitor oxygen requirement - Manage fever, pain, fluids, distress - Consider blood gas - If further deterioration, escalate to medical review immediately
Medical Review	
Deterioration Response Criteria - New or worsening increased work of breathing - Increased rate of apnoea/ bradycardia/desaturation episodes - New drop in SaO₂ consistently <85% despite appropriate first line interventions such as stimulation/suction/increase FiO₂ - New increase in FiO₂ by > 0.1 - A clear change in blood pressure (mean BP up or down by more than 5mmHg) - Any suspected seizures - Abnormal blood gas ($\leq$ pH 7.25) - Worsening or initial lactate >4mmol/L or base deficit >8 - PGL <2.6mmol/L - You (or a family member/carer) think that the infant requires medical review but they do not meet the above criteria	Actions Required - Contact registrar (phone 28806) with ISOBAR details, requesting review within 15 minutes - Commence continuous monitoring and record observations - If medical review not attended within 15 minutes escalate to Consultant - If ongoing deterioration initiate Code Blue Neonatal Emergency Call (via 55) Repeat any unexpected abnormal blood gas thought to be 'wrong' (e.g. poor collect, machine error) within 30 minutes

15.6. Appendix 6 ANZCOR A newborn life support algorithm

To be used in conjunction with [ANZCOR Guideline 13.1: Introduction to Resuscitation of the Newborn \(anzcor.org\)](https://www.anzcor.org/guidelines/13.1-introduction-to-resuscitation-of-the-newborn)

Resuscitation: Neonatal



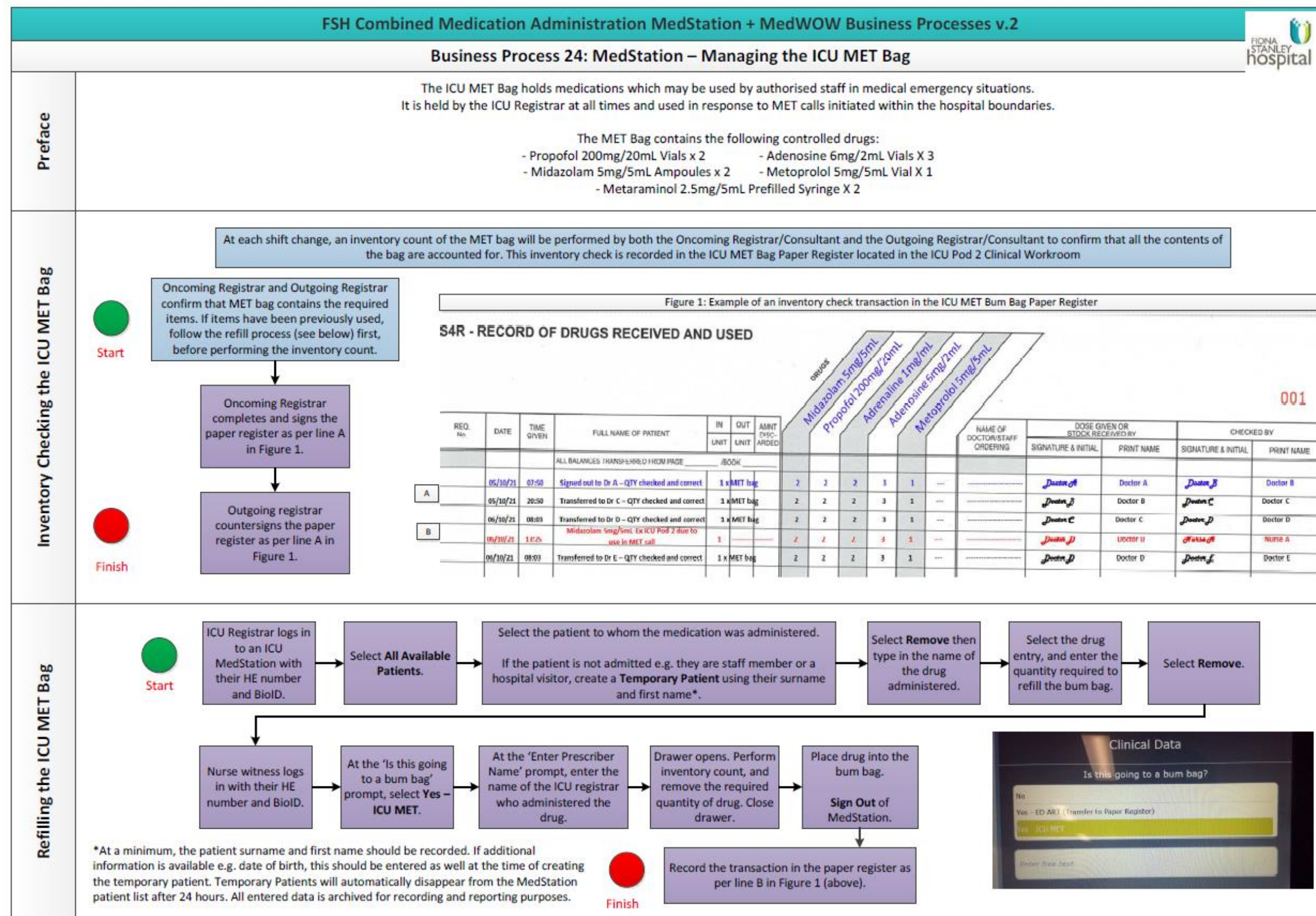
15.7. Appendix 7 ICU registrar MET bag contents list

ICU MET BAG CONTENTS	
DRUGS	Qty.
Propofol 200mg Vial	2
Midazolam 5mg Vial	2
Metaraminol 2.5mg/5ml Prefilled Syringe	2
Adrenaline 1:1000 (1mg/ml) Ampoule	2
Adenosine 6mg/2ml Vial	3
Metoprolol 5mg/5ml Vial	1
Potassium 5mmol/5ml Prefilled Syringe (in red sealed bag)	1
OTHER EQUIPMENT	
Bone Injection Gun	2
Laryngoscope Blade - Size 3	1
Laryngoscope Handle	1
10ml Syringe	3
20ml Syringe	1
19g Drawing up needle	4
NACL 0.9% Prefilled Flush	4

15.8. Appendix 8 MET SRN grab bag contents list

MET CNS BAG CONTENTS	
DRUGS	Qty.
2 x 5mg Metoprolol vials	2
3 x 6mg Adenosine vials	3
2 x 5mg metaraminol pre-filled syringe	2
GTN Spray	1
Co-phenylcaine spray and Nozzle	1
OTHER EQUIPMENT	
BIG IO	1
Coviden portable Spo2 Monitor	1
Forehead SpO2 sensor	1
Finger SpO2 sensor	1
Pen Torch	1
EtCo2 filter line	1
3-way Tap	1
Syringe infusion Line	1
Medium gloves	2
Gauze	2
18G 51mm Cannula	2
20G 51mm Cannula	2
12G 76mm angiocath	1
14G 76 mm angiocath	1
Airway Needle 19g	2

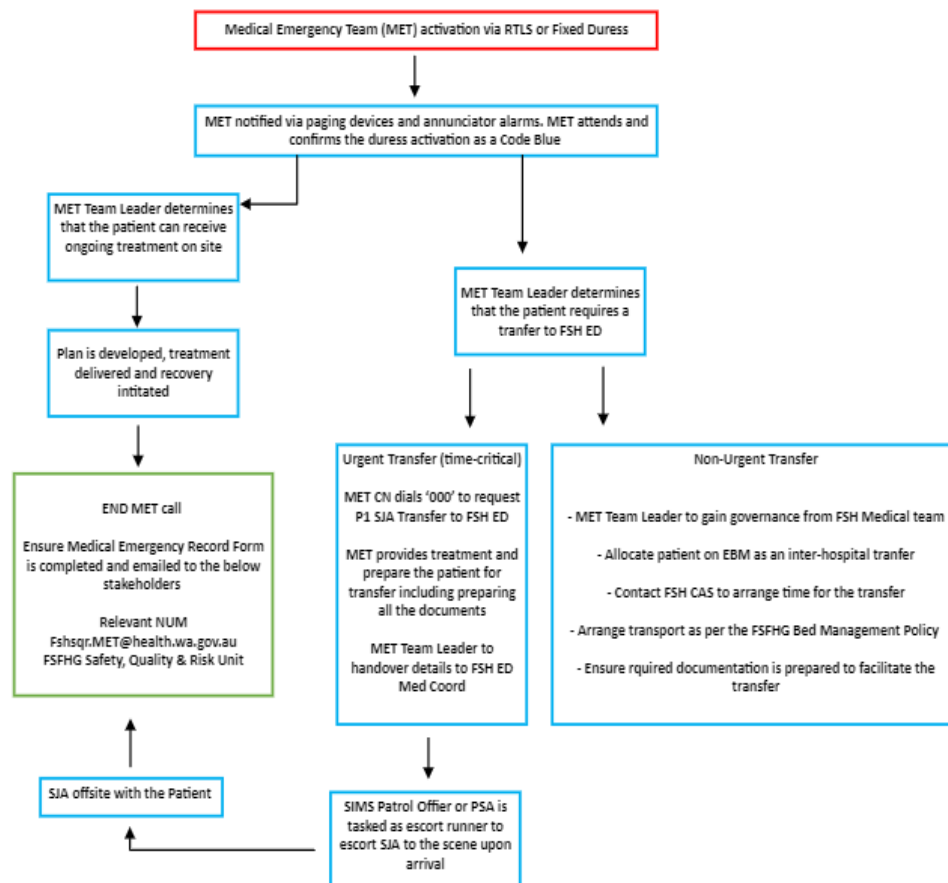
15.9. Appendix 9 FSH MET bag drug restocking procedure.



15.10. Appendix 10 FSH - Cockburn Health MET guidelines

15.10.1. Activating a Code Blue

- Staff are to activate their personal RTLS device and stay with the patient.
- The first MET team member present is to assess and advise staff to call '55' stating 'Code Blue' and request Ambulance if required. See FSH-CH Adult Activation Flowchart.



15.10.2. FSH-CH MET Response Team

The MET response team will be comprised of the following members of staff at the following times. Noting there may be differences across shift patterns.

- Team Leader – Psychiatric registrar (24/7)
- RMO (24/7)
- Allocated Clinical Nurse Specialist (Day shift only)
- HOOT Clinical Nurse Manager (Afternoon & Night shift)
- Allocated Clinical Nurse Ward 1, 2 & 3
- SIMS Patrol Officer/s
- Patient Support Services (Day & Afternoon Shift)

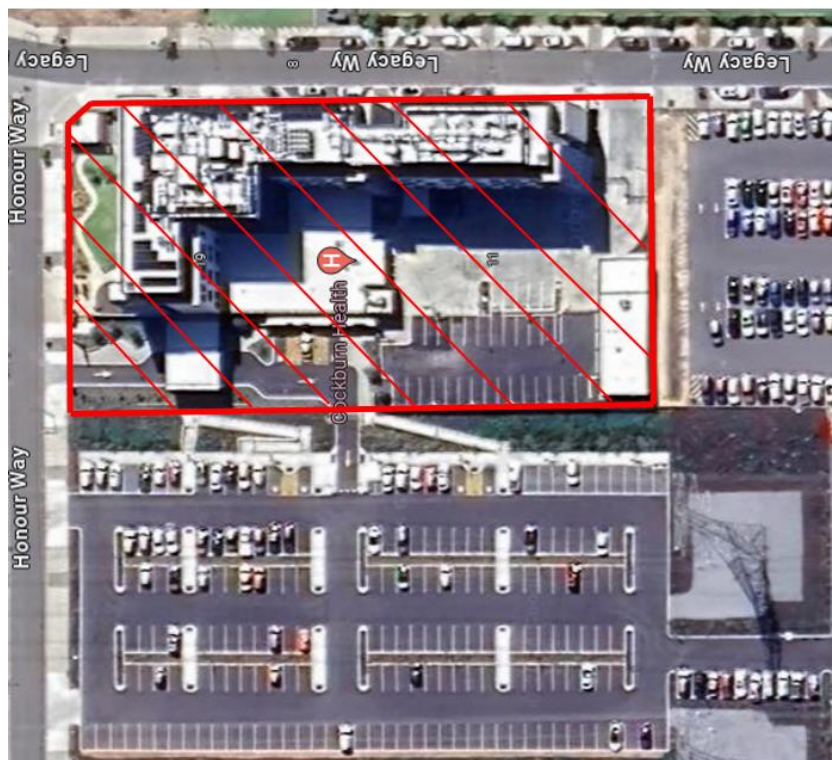
The MET Team Lead for the day must be allocated at the beginning of the morning SRN huddle. The huddle occurs in the ward huddle room.

For specific roles and responsibilities of the FSH CH MET members see FSH Cockburn Health – Code Blue Medical Emergency

Note: there is no airway specialist at FSH-CH

15.10.3. MET Boundaries at FSH CH

- The FSH-CH MET boundary is defined by the map below.
- For all other areas external to the FSH CH MET boundary first responders should provide assistance where able, or call an ambulance if necessary, on '000'.



15.10.4. FSH CH Requests from the public

FSH CH does not have an Emergency Department

- If a member of the public approaches administration staff requesting medical attention, they will inform the person that the nearest ED is at FSH and if necessary, provide directions/map.
- If the presenting individual meets Medical Emergency Team (MET) criteria, then a code blue must be activated (See FSH-CH Code Blue - Medical Emergency procedure).

15.10.5. FSH CH MET Governance

- On completion of a Code Blue medical emergency, the senior nurse on the MET is responsible for ensuring the accurate completion of the MET Record Form.
- A copy of the form is scanned and sent to Fshch.MET@health.wa.gov.au and the NUM of the area and recorded in the FSH CH MET Call spreadsheet.
- The original form is retained on the patient's medical record.
- MET call documentation is reviewed by the NUM of the area and the after-hours CNM to identify any failures in the recognition and response system.
- A quarterly report is compiled by FSH CH and tabled at the RRAD Committee identifying all MET calls for the facility resulting in an escalation of care and the recommendation and implementations post MET call. This includes:
 - Transfer to higher acuity ward
 - Transfer to another hospital
 - Death
- Recognised trends will be escalated to the FSFHG MET Coordinator for advice and support when required.
- Any cases that are identified as demonstrating issues are reported via a retrospective Datix Clinical Incident Management System (CIMS) submission.

15.10.6. Monitoring

- FSH CH NUMs, Afterhours CNM and SQR will complete regular review of clinical incident data via the Datix clinical incident management system.
- FSH CH NUMs, Afterhours CNM, SQR will work with the FSFHG RRAD Committee to undertake any regular audits and quality improvement activities set out by the committee, including gathering feedback from staff and patients/carers on the RRAD/Care call systems.